

Empirical Methods in Finance

Section 13 - Nonlinear Econometrics

Florian Ielpo, f.ielpo@lombardodier.com

HEC Lausanne

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Why nonlinearity matters in finance

The linear benchmark is useful - but incomplete

A large part of empirical finance starts from a linear conditional mean model:

$$y_t = x_t' \beta + u_t, \quad (1)$$

$$\mathbb{E}[u_t \mid x_t] = 0. \quad (2)$$

- It is interpretable: one slope measures one marginal effect.
- It is often the best first approximation, especially for expected returns.
- But many financial mechanisms are not linear: downside risk, volatility clustering, leverage effects, limits to arbitrage, crisis regimes and transaction costs.

Nonlinear econometrics asks whether the conditional distribution changes shape across states.

Three forms of nonlinearity

Let $\mathcal{F}_{t-1}$ be the information set.

$$\text{Mean nonlinearity: } \mathbb{E}[y_t \mid \mathcal{F}_{t-1}] = m(\mathcal{F}_{t-1}), \quad (3)$$

$$\text{Variance nonlinearity: } \text{Var}[y_t \mid \mathcal{F}_{t-1}] = h(\mathcal{F}_{t-1}), \quad (4)$$

$$\text{Distribution nonlinearity: } Q_{y_t}(\tau \mid \mathcal{F}_{t-1}) = q_\tau(\mathcal{F}_{t-1}). \quad (5)$$

- Mean nonlinearity is hard to detect in daily returns.
- Variance nonlinearity is usually more visible.
- Distribution nonlinearity matters for downside risk and portfolio construction.

A financial taxonomy of nonlinear effects

Economic mechanism	Econometric object	Example
Volatility clustering	Conditional variance	GARCH, stochastic volatility
Leverage effect	Asymmetric variance	EGARCH, GJR-GARCH
Crash risk	Tail quantiles	Quantile regression, EVT
State-dependent beta	Conditional slope	Downside beta, threshold beta
Regime dependence	Piecewise dynamics	TAR, SETAR, Markov switching
Unknown functional form	Conditional mean	Kernels, splines, ML

The main methodological choice is between interpretability and flexibility.

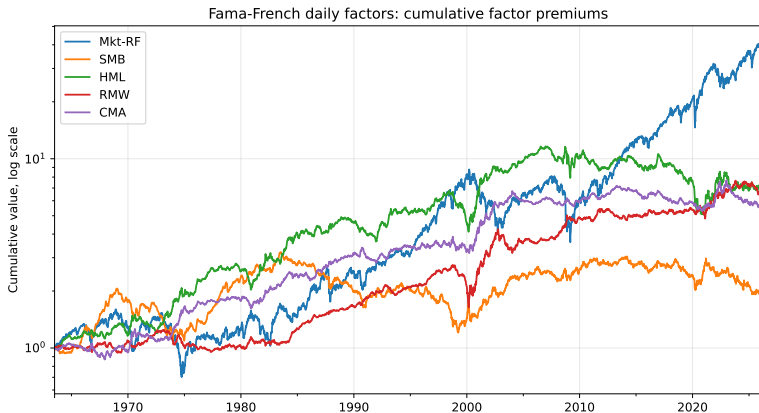
Data used in the application

The empirical application uses the Fama-French Research Data 5 Factors 2x3 at daily frequency.

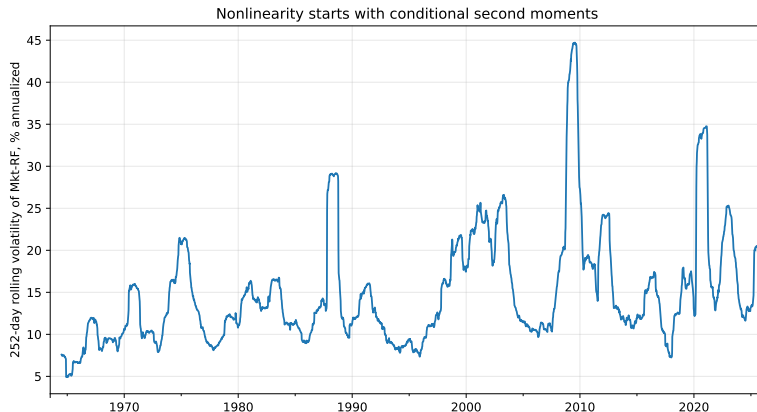
- Sample: to .
- Number of daily observations: .
- Factors: market excess return, size, value, profitability and investment.
- Market excess return annualized mean: %; annualized volatility: %.
- Market excess return skewness: ; kurtosis: .

The data are used to illustrate models, not to claim a tradable signal.

Fama-French factors: cumulative premiums



Volatility clustering is the simplest visual evidence



Nonlinear estimation and inference

From linear to nonlinear regression

A nonlinear regression model is written as:

$$y_t = g(x_t, \theta_0) + u_t, \quad (6)$$

$$\mathbb{E}[u_t \mid x_t] = 0, \quad (7)$$

where $g(\cdot)$ is known but nonlinear in parameters or variables.

- Nonlinear in variables but linear in parameters:

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 x_t^2 + u_t. \quad (8)$$

- Nonlinear in parameters:

$$y_t = \theta_1 \exp(\theta_2 x_t) + u_t. \quad (9)$$

The second case requires nonlinear least squares or maximum likelihood.

The nonlinear least squares estimator solves:

$$\hat{\theta} = \arg \min_{\theta \in \Theta} Q_T(\theta), \quad (10)$$

$$Q_T(\theta) = \frac{1}{T} \sum_{t=1}^T (y_t - g(x_t, \theta))^2. \quad (11)$$

Let $G_t(\theta) = \partial g(x_t, \theta) / \partial \theta$. The first order condition is:

$$\frac{\partial Q_T(\theta)}{\partial \theta} = -\frac{2}{T} \sum_{t=1}^T G_t(\theta) (y_t - g(x_t, \theta)) = 0. \quad (12)$$

Unlike OLS, this condition is typically nonlinear in θ and must be solved numerically.

Asymptotic distribution of NLS

Under standard regularity conditions:

$$\sqrt{T}(\hat{\theta} - \theta_0) \rightarrow_d N(0, A^{-1}BA^{-1}), \quad (13)$$

with

$$A = \text{plim} \frac{1}{T} \sum_{t=1}^T G_t(\theta_0) G_t(\theta_0)', \quad (14)$$

$$B = \text{plim} \frac{1}{T} \sum_{t=1}^T G_t(\theta_0) G_t(\theta_0)' u_t^2. \quad (15)$$

If errors are conditionally homoskedastic with variance σ^2 :

$$\sqrt{T}(\hat{\theta} - \theta_0) \rightarrow_d N(0, \sigma^2 A^{-1}). \quad (16)$$

In finance, the robust sandwich form is usually safer.

Numerical estimation: why starting values matter

Most nonlinear estimators are computed by iterations:

$$\theta^{(j+1)} = \theta^{(j)} - \lambda_j H(\theta^{(j)})^{-1} \nabla Q_T(\theta^{(j)}), \quad (17)$$

where $H(\cdot)$ is a Hessian or an approximation to it.

- The objective can be non-convex.
- Several local optima can coexist.
- Parameters may be weakly identified in some regimes.
- Scaling matters: returns in decimals are easier to optimize than returns in percent.

Good applied practice: try several starting values, inspect residuals, compare with a linear benchmark.

Nonlinear maximum likelihood

For a conditional density $f(y_t | \mathcal{F}_{t-1}; \theta)$, maximum likelihood solves:

$$\hat{\theta} = \arg \max_{\theta \in \Theta} \ell_T(\theta), \quad (18)$$

$$\ell_T(\theta) = \sum_{t=1}^T \log f(y_t | \mathcal{F}_{t-1}; \theta). \quad (19)$$

A common Gaussian example is:

$$y_t = m_t(\theta) + \varepsilon_t, \quad (20)$$

$$\varepsilon_t | \mathcal{F}_{t-1} \sim N(0, h_t(\theta)), \quad (21)$$

$$\ell_T(\theta) = -\frac{1}{2} \sum_{t=1}^T \left[\log h_t(\theta) + \frac{(y_t - m_t(\theta))^2}{h_t(\theta)} \right] + C. \quad (22)$$

This is the backbone of GARCH and many regime models.

Model selection: in-sample fit is not enough

Nonlinear models almost always fit better in-sample. That is not a sufficient reason to use them.

$$AIC = -2\ell_T(\hat{\theta}) + 2k, \quad (23)$$

$$BIC = -2\ell_T(\hat{\theta}) + k \log(T), \quad (24)$$

where k is the number of estimated parameters.

- AIC is more tolerant of complexity.
- BIC penalizes complexity more strongly.
- Out-of-sample evaluation is essential when the model is used for forecasting or allocation.

The key question is not “does it fit better?” but “does it explain economically meaningful state dependence?”

Testing for nonlinearity

A disciplined testing workflow

A useful sequence is:

1. Estimate a linear benchmark.
2. Inspect residuals and conditional moments.
3. Apply specification tests.
4. Estimate a parsimonious nonlinear model.
5. Check whether the nonlinear term is stable and economically interpretable.
6. Validate out-of-sample if the model will be used for decisions.

In finance, a test can reject linearity because of conditional heteroskedasticity rather than true conditional mean nonlinearity. Always separate the two.

RESET as a first specification check

Start with the linear model:

$$y_t = x_t' \beta + u_t. \quad (25)$$

Let $\hat{y}_t = x_t' \hat{\beta}$. The Ramsey RESET regression is:

$$y_t = x_t' \beta + \gamma_2 \hat{y}_t^2 + \gamma_3 \hat{y}_t^3 + e_t. \quad (26)$$

The null is:

$$H_0 : \gamma_2 = \gamma_3 = 0. \quad (27)$$

- Advantage: simple and fast.
- Limitation: it does not tell which nonlinear model to estimate.

BDS test: residual dependence beyond linear correlation

The BDS test checks whether a series is iid by comparing correlation integrals in different embedding dimensions.

$$C_m(\varepsilon) = \frac{2}{T_m(T_m - 1)} \sum_{i < j} \mathbb{1}(\|X_i^m - X_j^m\| < \varepsilon), \quad (28)$$

where $X_i^m = (x_i, \dots, x_{i+m-1})$. Under iid:

$$C_m(\varepsilon) \approx C_1(\varepsilon)^m. \quad (29)$$

The statistic is built from:

$$W_{m,T}(\varepsilon) = \frac{\sqrt{T}[C_m(\varepsilon) - C_1(\varepsilon)^m]}{\hat{\sigma}_m(\varepsilon)}. \quad (30)$$

Use it on residuals, not on raw returns, after fitting the linear benchmark.

Tsay test for threshold nonlinearity

Tsay's arranged autoregression test is designed for threshold autoregressive dynamics.

$$y_t = \alpha + \sum_{j=1}^p \phi_j y_{t-j} + u_t. \quad (31)$$

Choose a delay variable $q_t = y_{t-d}$. Sort observations by q_t , estimate recursive regressions, and test whether residuals are correlated with regressors arranged by threshold order.

- Null: one linear autoregression is sufficient.
- Alternative: autoregressive coefficients change when q_t crosses a threshold.
- Interpretation: useful before estimating SETAR or TAR models.

Hansen's threshold test: why the usual Wald test fails

In a threshold model, the threshold τ is not identified under the null of linearity.

$$y_t = x_t' \beta + x_t' \delta \mathbb{1}(q_t \leq \tau) + u_t. \quad (32)$$

Under the null:

$$H_0 : \delta = 0, \quad (33)$$

any value of τ gives the same model. The Wald statistic is therefore non-standard. A practical solution is the sup-Wald statistic:

$$\sup_{\tau \in \mathcal{T}} W_T(\tau), \quad (34)$$

with critical values obtained by bootstrap.

A logistic smooth-transition model can be written as:

$$y_t = x_t' \beta + x_t' \delta G(q_t; \gamma, c) + u_t, \quad (35)$$

where

$$G(q_t; \gamma, c) = (1 + \exp[-\gamma(q_t - c)])^{-1}. \quad (36)$$

Under linearity, $\gamma = 0$, which creates an identification problem. Terasvirta's LM test uses a Taylor expansion that leads to an auxiliary regression with powers and interactions of q_t . It is a structured way to test gradual rather than abrupt nonlinearity.

Testing nonlinearity: what to remember

Test	Main target	Key limitation
RESET	Generic misspecification	Not model-specific
BDS	Residual dependence	Sensitive to heteroskedasticity
Tsay	Threshold autoregression	Requires ordering variable
Hansen sup-Wald	Threshold effects	Needs bootstrap
Terasvirta LM	Smooth transition	Approximation-based
ARCH LM	Conditional variance	Not mean nonlinearity

A rejection is a diagnosis. It is not yet an economic model.

Threshold and smooth-transition models

A two-regime TAR model is:

$$y_t = \alpha_1 + \sum_{j=1}^p \phi_{1j} y_{t-j} + u_{1t}, \quad q_t \leq \tau, \quad (37)$$

$$y_t = \alpha_2 + \sum_{j=1}^p \phi_{2j} y_{t-j} + u_{2t}, \quad q_t > \tau. \quad (38)$$

- q_t is the transition variable.
- τ is the threshold.
- Regime coefficients can differ in mean reversion, volatility and persistence.

When $q_t = y_{t-d}$, the model is a SETAR model.

TAR estimation by grid search

For each candidate threshold τ , estimate by OLS:

$$y_t = x_t' \beta + x_t' \delta \mathbb{1}(q_t \leq \tau) + u_t. \quad (39)$$

Then select τ by minimizing the sum of squared residuals:

$$\hat{\tau} = \arg \min_{\tau \in \mathcal{T}} SSR(\tau), \quad (40)$$

where

$$SSR(\tau) = \sum_{t=1}^T \hat{u}_t(\tau)^2. \quad (41)$$

In practice, trim the first and last 10% or 15% of observations to avoid regimes with too few data points.

A smooth transition regression replaces the hard indicator by a continuous transition function:

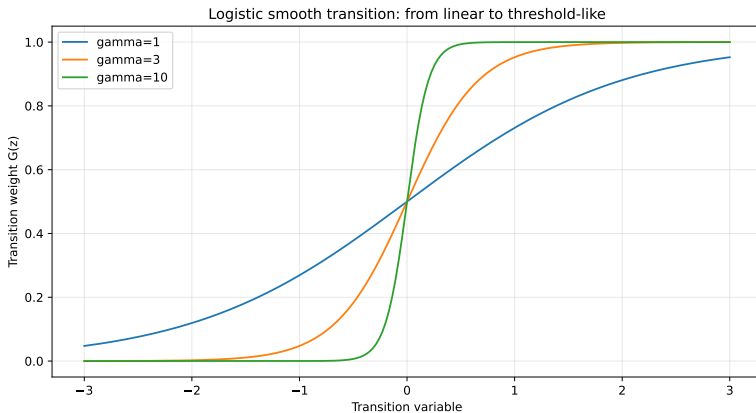
$$y_t = x_t' \beta + x_t' \delta G(q_t; \gamma, c) + u_t. \quad (42)$$

For the logistic transition:

$$G(q_t; \gamma, c) = (1 + \exp[-\gamma(q_t - c)])^{-1}. \quad (43)$$

- c is the location of the transition.
- γ controls the speed of transition.
- As $\gamma \rightarrow \infty$, the model becomes threshold-like.

Logistic transition function



Threshold versus smooth transition

Feature	Threshold model	Smooth transition model
State change	Abrupt	Gradual
Parameters	OLS for fixed threshold	Nonlinear least squares
Interpretation	Crisis switch	Progressive stress
Identification	Threshold under null	Speed under null
Best use	Clear regimes	Transition phases

In financial markets, abrupt models are useful for crashes, while smooth models are useful for changing monetary or liquidity conditions.

Application: threshold search on market returns

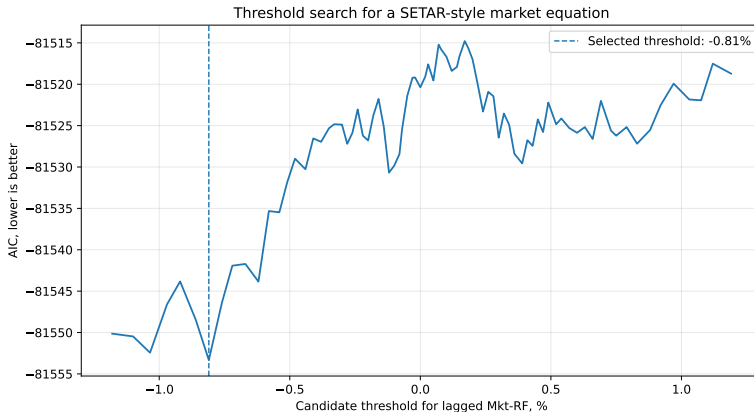
We estimate a SETAR-style equation on daily market excess returns from 1990 onward:

$$r_t = \alpha + \phi r_{t-1} + \delta_0 \mathbb{1}(r_{t-1} \leq \tau) + \delta_1 r_{t-1} \mathbb{1}(r_{t-1} \leq \tau) + u_t. \quad (44)$$

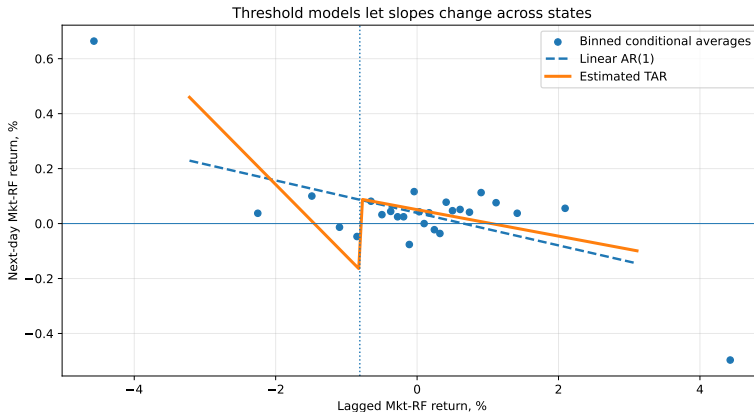
- Transition variable: lagged market excess return.
- Candidate thresholds: 10th to 90th percentile of lagged returns.
- Selection criterion: AIC from the Gaussian SSR approximation.

Selected threshold: %.

Threshold search: AIC across candidate thresholds



Estimated threshold fit



Application result: what changes across states?

The linear AR(1) slope is:

$$\hat{\phi}_{linear} = . \quad (45)$$

The TAR slopes are:

$$\hat{\phi}_{low} = , \quad (46)$$

$$\hat{\phi}_{high} = . \quad (47)$$

- After strongly negative days, the slope is more negative.
- This suggests stronger short-run reversal after stress events.
- The interpretation is economic only if it remains stable across samples and survives transaction costs.

Non-parametric estimation

The non-parametric problem

Assume the conditional mean is unknown:

$$y_t = m(x_t) + u_t, \quad \mathbb{E}[u_t \mid x_t] = 0. \quad (48)$$

Linear regression imposes:

$$m(x) = x' \beta. \quad (49)$$

Non-parametric regression estimates $m(\cdot)$ directly from the data.

- Advantage: fewer functional form assumptions.
- Cost: slower convergence and strong dependence on smoothing choices.
- Best use: exploration, diagnostics and low-dimensional relationships.

The Nadaraya-Watson estimator is:

$$\hat{m}(x) = \frac{\sum_{t=1}^T K\left(\frac{x_t - x}{h}\right) y_t}{\sum_{t=1}^T K\left(\frac{x_t - x}{h}\right)}. \quad (50)$$

It can be written as a weighted average:

$$\hat{m}(x) = \sum_{t=1}^T w_t(x) y_t, \quad (51)$$

where

$$w_t(x) = \frac{K\left(\frac{x_t - x}{h}\right)}{\sum_{s=1}^T K\left(\frac{x_s - x}{h}\right)}. \quad (52)$$

Nearby observations receive high weights; distant observations receive low weights.

Bandwidth is the key tuning parameter

For a univariate kernel estimator:

$$\text{Bias} [\hat{m}(x)] \approx C_1 h^2, \quad (53)$$

$$\text{Var} [\hat{m}(x)] \approx \frac{C_2}{Th}. \quad (54)$$

The bandwidth controls the bias-variance trade-off:

- Small h : flexible fit, high variance.
- Large h : smooth fit, high bias.

A rule-of-thumb bandwidth is:

$$h = 1.06 \hat{\sigma}_x T^{-1/5}. \quad (55)$$

In the application, this gives about 10 basis points for lagged market returns.

Local linear regression

Nadaraya-Watson can be biased near boundaries. Local linear regression solves, for each point x :

$$(\hat{a}(x), \hat{b}(x)) = \arg \min_{a,b} \sum_{t=1}^T K\left(\frac{x_t - x}{h}\right) [y_t - a - b(x_t - x)]^2. \quad (56)$$

The local estimate is:

$$\hat{m}(x) = \hat{a}(x). \quad (57)$$

- It reduces boundary bias.
- It is often preferred for smooth conditional means.
- It remains sensitive to bandwidth and outliers.

A kernel can also estimate the full density:

$$\hat{f}(x) = \frac{1}{Th} \sum_{t=1}^T K\left(\frac{x_t - x}{h}\right). \quad (58)$$

This is the Rosenblatt-Parzen density estimator.

- It is useful to inspect fat tails and multimodality.
- It can be applied to raw returns or standardized residuals.
- A model that removes volatility clustering should make residual density more stable.

The curse of dimensionality

In dimension d , a kernel estimator uses a neighborhood volume proportional to h^d :

$$\text{Var} [\hat{m}(x)] \propto \frac{1}{Th^d}. \quad (59)$$

To keep variance controlled as d rises, one needs many more observations.

- Non-parametric methods work best for one to three explanatory variables.
- High-dimensional finance usually requires structure: additive models, splines, trees, shrinkage or economic restrictions.
- More flexibility is not free: it is paid for in data requirements.

An alternative is to approximate $m(x)$ by basis functions:

$$m(x) \approx \sum_{k=1}^K \beta_k p_k(x). \quad (60)$$

Examples:

- Polynomials: $1, x, x^2, x^3$.
- Splines: piecewise polynomials connected at knots.
- Fourier bases: sine and cosine functions.

The estimator is linear in β , but the fitted relationship can be highly nonlinear in x .

Machine learning as nonlinear econometrics

Many machine learning methods estimate:

$$y_t = m(x_t) + u_t, \quad (61)$$

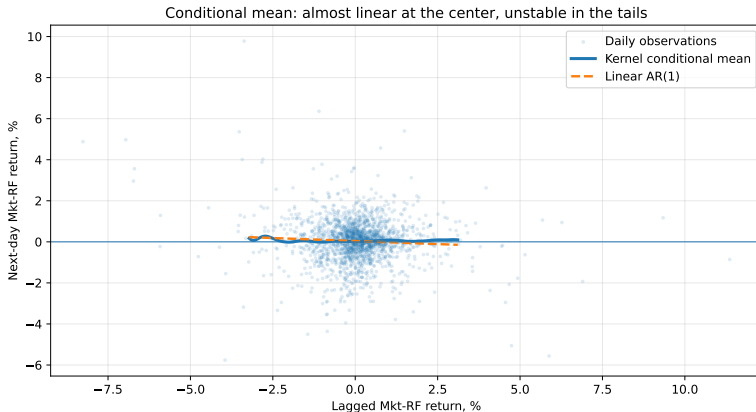
with a flexible $m(\cdot)$.

Method	Economic analogy	Main risk
Trees	Threshold models	Instability
Random forests	Averaged thresholds	Low interpretability
Boosting	Sequential nonlinear fit	Overfitting
Neural networks	Sieve approximation	Hard identification

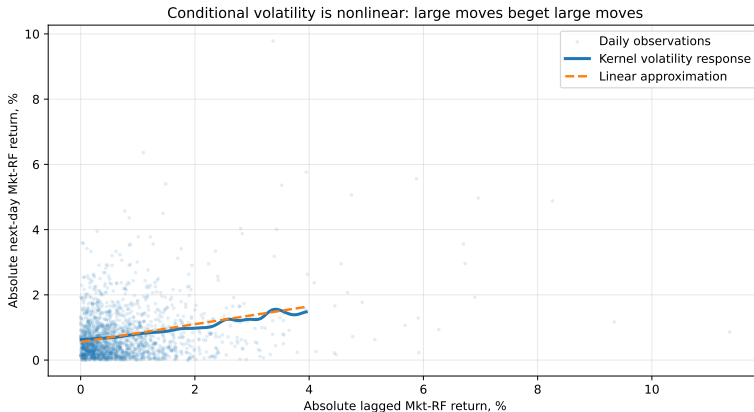
The econometric discipline remains the same: define the loss, validate out-of-sample, interpret economically.

Application: Fama-French five factors

Kernel conditional mean for market returns



Kernel volatility response



Conditional beta as a nonlinear factor exposure

A standard beta is:

$$r_{i,t} = \alpha_i + \beta_i r_{m,t} + u_{i,t}. \quad (62)$$

A downside beta lets the exposure change in bad market states:

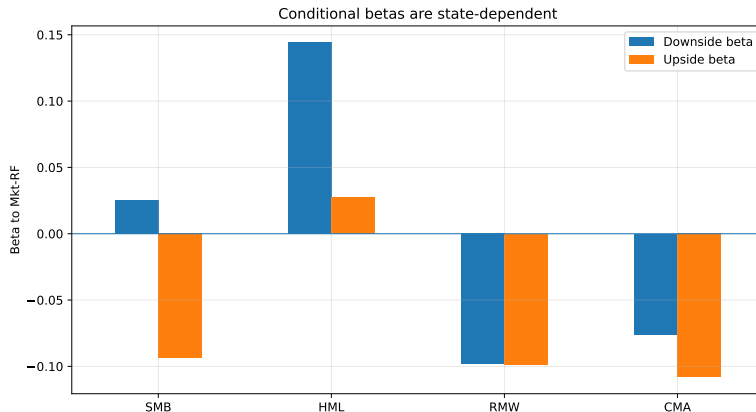
$$r_{i,t} = \alpha_i + \beta_i r_{m,t} + \gamma_i r_{m,t} \mathbb{1}(r_{m,t} \leq q_{20}) + u_{i,t}. \quad (63)$$

Then:

$$\beta_i^{\text{down}} = \beta_i + \gamma_i. \quad (64)$$

This is a simple nonlinear factor model: the marginal exposure is state-dependent.

Downside and upside betas of Fama-French factors



Quantile regression

OLS focuses on the conditional mean:

$$\hat{\beta}^{OLS} = \arg \min_{\beta} \sum_{t=1}^T (y_t - x_t' \beta)^2. \quad (65)$$

Quantile regression estimates a conditional quantile:

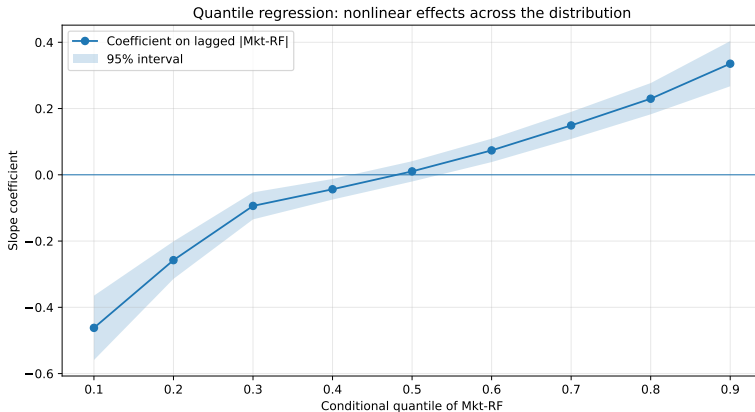
$$\hat{\beta}(\tau) = \arg \min_{\beta} \sum_{t=1}^T \rho_{\tau}(y_t - x_t' \beta), \quad (66)$$

where

$$\rho_{\tau}(u) = u(\tau - \mathbb{1}(u < 0)). \quad (67)$$

It is a natural tool when downside and upside states have different slopes.

Quantile regression: the effect changes across the distribution



How to interpret the application

- The conditional mean of daily market returns remains hard to predict.
- The conditional variance response is much more visible: large lagged moves are associated with larger next-day absolute returns.
- A simple threshold model finds a stress threshold around %.
- Conditional betas show that factor exposures can differ in downside and upside market states.
- The practical lesson is not “use the most nonlinear model”; it is “diagnose which conditional moment is nonlinear.”

Empirical workflow for a student project

1. Start with a linear benchmark: OLS, AR, VAR or factor regression.
2. Test residuals: autocorrelation, ARCH effects, RESET or BDS.
3. Plot non-parametric diagnostics: kernel mean, kernel volatility, conditional beta.
4. Estimate one interpretable nonlinear model: threshold, smooth transition or quantile regression.
5. Compare fit with AIC/BIC and out-of-sample loss.
6. Conclude with economics: which state changes, why, and what it implies for risk?

- Confusing nonlinear fit with economic predictability.
- Estimating too many regimes with too few observations per regime.
- Forgetting that threshold critical values are non-standard.
- Ignoring conditional heteroskedasticity when testing mean nonlinearity.
- Using a non-parametric model in high dimension without enough data.
- Presenting a black-box model without a benchmark and without economic interpretation.

- Nonlinear econometrics is about state-dependent conditional distributions.
- Finance offers strong nonlinearities in volatility, tails and betas; expected-return nonlinearities are harder to exploit.
- Parametric nonlinear models are interpretable but restrictive.
- Non-parametric models are flexible but data-hungry.
- A good empirical strategy combines both: non-parametric diagnostics first, parsimonious nonlinear model second.

The objective is not complexity. The objective is a better description of financial risk across states.

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